Robo's Turning Numbers

Imaginary Numbers and Euler's Equation

October 31, 2026 · Nathaniel

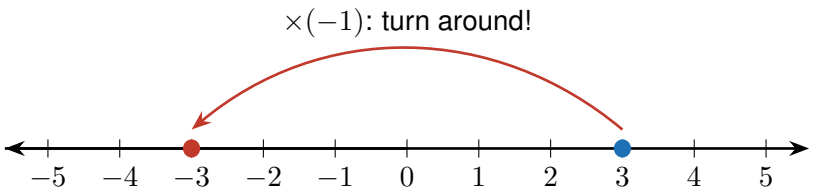
Name: _____ Date: _____

The story

Robo the Robot lives on a number line. Robo stands on a number and faces the way the numbers grow. Robo knows two tricks: **walking** and **turning**.

One day a mathematician asks Robo a very tricky riddle. To solve it, Robo will discover a whole new kind of number, and one of the most famous equations in all of math!

Trick 1: times -1 means "turn around"



Multiplying by -1 is like Robo doing a **half turn**: Robo spins around to face the other way, and the number jumps to the other side of 0. So $3 \times (-1) = -3$.

Question 1. $4 \times (-1) =$ _____

Question 2. Turn around *two* times: $2 \times (-1) \times (-1) =$ _____

Which way is Robo facing now? _____

Try it! Stand up and face the board. Do a half turn. Do another half turn. Which way are you facing?

The tricky riddle

The mathematician asks: *"Is there a number that, if you multiply by it **two** times, makes you turn around?"*

Question 3. Robo tries. Fill in the blanks.

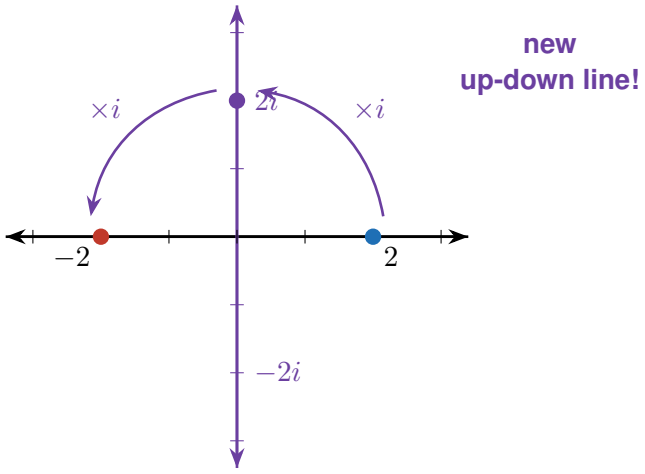
$1 \times 1 =$ _____ Did Robo turn around? _____

$(-1) \times (-1) =$ _____ Did Robo turn around? _____

Nothing on the number line works! Robo needs a brand new idea.

Trick 2: the quarter turn, i

Try it! Do a half turn using **two quarter turns**: turn left a little (a quarter turn), then turn left again. You are facing backwards. It worked!



Mathematicians invented a new number called i. Multiplying by i is a **quarter turn** to the left. It makes Robo walk off the number line onto a new **up-down line**. Numbers on the up-down line are called **imaginary numbers**.
Two quarter turns make a half turn, so $i \times i = -1$. Riddle solved!

The turning pattern

Robo starts at 1 and keeps multiplying by i. Each time, Robo does one more quarter turn. Use the picture on the page before to fill in where Robo lands.

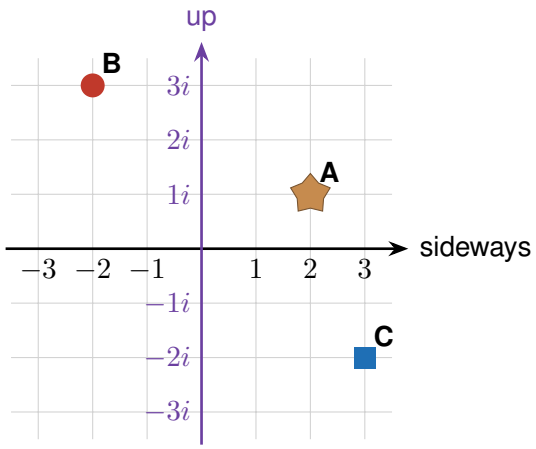
Quarter turns	Robo is at	Pointing
0	1	right
1	i	up
2		
3		
4		

Question 4. How many quarter turns bring Robo all the way back to 1? _____

Question 5. The pattern repeats! Where is Robo after 8 quarter turns? (Count by 4s.) _____

Robo's treasure map

Now numbers can go **sideways** and **up**! A number like $2 + 1i$ is an address: walk 2 steps right, then 1 step up.



Question 6. The star **A** is at $2 + 1i$. Write the address of the others.

B (red circle): _____ $+$ _____ i **C** (blue square): _____ $+$ _____ i

Question 7. Draw an **X** on the map at $1 + 3i$.

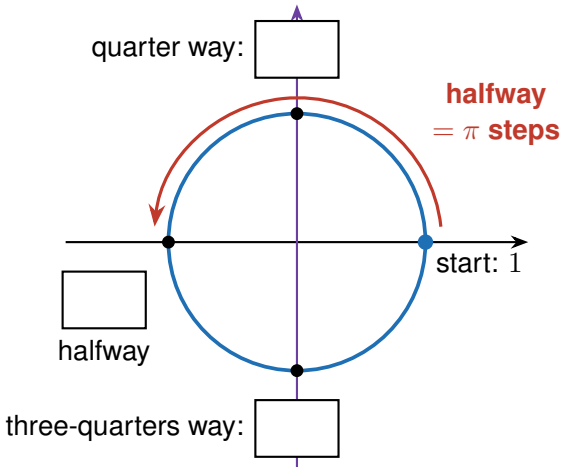
Robo walks around a circle

Try it! Wrap a string once around a cup. Then see how many times the cup's width fits along the string.

Question 8. About how many cup-widths long was the string? _____

It is always a little more than 3, for *every* circle! That number is called **pi**, written π. It is about 3.14.
If a circle is 2 steps wide, the path all the way around is 2π steps long, so **halfway around is exactly π steps**.

Robo starts at 1 and walks around a circle that is 2 steps wide. Fill in the boxes with where Robo lands: 1, i, -1, or $-i$.



Euler's equation

How a scientist writes "walk around the circle"

A mathematician named **Leonhard Euler** (say "OY-ler") found a way to write Robo's circle walk. There is a special number called e, about 2.72. When you write

$$e^{i \times \text{steps}}$$

it means: **start at 1 and walk that many steps around the circle.**

Question 9. Walk π steps (halfway around). Where does Robo land?

$$e^{i\pi} = _____$$

Question 10. Now add 1 to both sides. $-1 + 1 =$ _____. So:

$$e^{i\pi} + 1 = _____$$

Five famous numbers in one equation

$$e^{i\pi} + 1 = 0$$

Many mathematicians say this is the **most beautiful equation** in all of math, because it uses five famous numbers at once.

Question 11. Draw a line from each number to what it means.

0 •	• The quarter-turn number
1 •	• Nothing at all
i •	• The special number for Euler's circle walk, about 2.72
π •	• Where Robo starts
e •	• How far halfway around the circle is, about 3.14

Draw it!

Draw Robo walking halfway around the circle, from 1 to -1. Write $e^{i\pi} = -1$ next to your picture.

For teachers: multiplying by i rotates the complex plane by $90°$ counterclockwise, which is why $i^2 = -1$ reads as two quarter turns making a half turn. Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ places $e^{i\theta}$ on the unit circle at arc length θ from 1 (counterclockwise), so $\theta = \pi$ lands on -1. Robo's "circle 2 steps wide" is the unit circle (radius 1), whose circumference is 2π. Why e in particular is the base that turns "growth" into "rotation" is a calculus fact ($\frac{d}{d\theta}e^{i\theta} = i\,e^{i\theta}$: the velocity is always a quarter turn from the position) and is beyond this worksheet.

Answer key

Question	Answer
1	-4
2	2; facing forward again (the way the numbers grow)
3	$1 \times 1 = 1$, no; $(-1) \times (-1) = 1$, no
Turning table	2 turns: -1, left 3 turns: $-i$, down 4 turns: 1, right
4	4 quarter turns
5	1 (8 is two groups of 4)
6	B is $-2 + 3i$; C is $3 + (-2)i$, which is also written $3 - 2i$
7	X one step right, three steps up
8	A little more than 3 (about 3.14 if measured carefully)
Circle boxes	Quarter way: i; halfway: -1; three-quarters: $-i$
9	-1
10	0; so $e^{i\pi} + 1 = 0$
11	0: nothing at all; 1: where Robo starts; i: the quarter-turn number; π: halfway around, about 3.14; e: Euler's special number, about 2.72